SolarAlign Field Data Worksheet

Collect the roof, solar, utility, and project information needed for a transparent timing and value analysis.

Write down the source for every answer. Estimates are useful, but label them as estimates. Attach roof reports, proposals, utility bills, and production studies where available.

Project / building: _____

Address: _____

Collector and date: _____

PROJECT AND BUILDING CONTEXT

These details identify the project and help the team verify that later estimates apply to the right building. They do not directly change the current calculation unless noted.

Project or building name

PROJECT CONTEXT

What to collect: A clear name for this analysis.
Why it matters: Keeps reports, source documents, and calculation results tied to the same project.
Suggested source: Owner, facility team, or project file

Value / observation: _____

Source document, page, or person: _____

Building address

PROJECT CONTEXT

What to collect: Street address, city, state, and ZIP code.
Why it matters: Location affects solar resource, utility territory, rates, permitting, and incentive eligibility, even though this version accepts annual production as an input.
Suggested source: Property records or utility bill

Value / observation: _____

Source document, page, or person: _____

Project / facilities contact

PROJECT CONTEXT

What to collect: Name, role, phone, or email for the person coordinating information.
Why it matters: Makes it possible to resolve missing roof, utility, and proposal facts before a decision is made.
Suggested source: Project team

Value / observation: _____

Source document, page, or person: _____

Site visit or report date [date]

PROJECT CONTEXT

What to collect: The date the roof or property information was observed or reported.
Why it matters: Roof condition and cost estimates age over time; recording the date helps reviewers judge whether information is current.
Suggested source: Roof report or field visit

Value / observation: _____

Source document, page, or person: _____

Field data worksheet - continued

Usable roof area [square feet]

PROJECT CONTEXT

What to collect: Approximate unobstructed roof area potentially available for solar.
Why it matters: Useful roof area helps a designer estimate feasible system size. It is contextual here because the current engine uses annual kWh from a production study.
Suggested source: Roof plan, aerial measurement, or solar proposal

Value / observation: _____

Source document, page, or person: _____

Proposed solar system size [kW DC]

PROJECT CONTEXT

What to collect: The direct-current nameplate capacity of the proposed array.
Why it matters: System size connects roof area, project cost, and energy production and is essential when validating a production or cost estimate.
Suggested source: Solar proposal or preliminary design

Value / observation: _____

Source document, page, or person: _____

Roof type / membrane

PROJECT CONTEXT

What to collect: For example: TPO, EPDM, modified bitumen, metal, or built-up roof.
Why it matters: Roof type affects attachment method, compatibility, warranty coordination, installation effort, and future removal work.
Suggested source: Roof report, warranty, or maintenance records

Value / observation: _____

Source document, page, or person: _____

Estimated remaining roof life [years or range]

PROJECT CONTEXT

What to collect: Record a range if the report gives one; do not turn a range into an exact fact.
Why it matters: Helps the team choose a roof-replacement scenario and decide whether solar should wait, proceed now, or be coordinated with reroofing.
Suggested source: Qualified roof assessment

Value / observation: _____

Source document, page, or person: _____

Inspection limitations

PROJECT CONTEXT

What to collect: Record areas not accessed, tests not performed, concealed conditions, and other limits on the available roof evidence.
Why it matters: Makes uncertainty visible so a preliminary financial result is not mistaken for a complete roof or structural assessment.
Suggested source: Roof or property condition report

Value / observation: _____

Source document, page, or person: _____

Structural capacity review status

PROJECT CONTEXT

What to collect: State whether rooftop solar loading has been assessed, by whom, and when.
Why it matters: Financial results do not prove physical feasibility. A qualified structural review is needed before installation.
Suggested source: Structural engineer or property condition report

Value / observation: _____

Source document, page, or person: _____

Field data worksheet - continued

ROOF AND SOLAR COSTS

These figures determine when cash is spent and whether coordinating roof work with solar avoids future disruption.

Selected roof replacement timing [years from today]

REQUIRED FOR MODEL

What to collect: The planning year used for comparison: 0 means now, 3 means three years from now.
Why it matters: Sets the roof-only baseline, the installation date for the roof-then-solar plan, and the year of panel removal and downtime if solar goes first.
Suggested source: Roof assessment plus project-team planning decision

Value / observation: _____

Source document, page, or person: _____

Roof replacement cost [USD]

REQUIRED FOR MODEL

What to collect: Expected total cost for one roof replacement in consistent dollars.
Why it matters: The engine charges this cost once in every plan. Paying it earlier changes its present value and can change which timing plan is viable.
Suggested source: Qualified contractor budget or capital plan

Value / observation: _____

Source document, page, or person: _____

Solar removal and reinstallation cost [USD]

REQUIRED FOR MODEL

What to collect: Cost to safely detach, store, and reinstall the array during future roof work.
Why it matters: Applies when solar is installed before the roof is replaced and directly reduces that plan's project-side NPV.
Suggested source: Solar installer or qualified contractor allowance

Value / observation: _____

Source document, page, or person: _____

Solar installation cost [USD]

REQUIRED FOR MODEL

What to collect: Total installed capital cost before incentives unless the source clearly states otherwise.
Why it matters: This is the largest upfront solar cash outflow and strongly affects whether any timing plan creates positive long-term value.
Suggested source: Solar proposal with scope and exclusions

Value / observation: _____

Source document, page, or person: _____

Annual operations and maintenance cost [USD per year]

REQUIRED FOR MODEL

What to collect: Expected yearly inspection, monitoring, service, and maintenance cost.
Why it matters: The engine subtracts this in each operating year, so omitting it would overstate project value.
Suggested source: Service proposal, asset manager, or benchmark

Value / observation: _____

Source document, page, or person: _____

Field data worksheet - continued

Production lost during roof work [percent of that year]

REQUIRED FOR MODEL

What to collect: Estimated share of annual production lost while panels are removed and reinstalled.
Why it matters: Reduces electricity value in the roof-work year for the solar-first plan.
Suggested source: Roof/solar construction schedule

Value / observation: _____

Source document, page, or person: _____

ENERGY PRODUCTION AND VALUE

These inputs describe how much electricity the array is expected to make and what that electricity is worth over time.

First-year annual solar production [kWh per year]

REQUIRED FOR MODEL

What to collect: Expected alternating-current electricity in the first full operating year.
Why it matters: The engine multiplies production by electricity value to calculate yearly benefits. Use a site-specific study such as PVWatts or a proposal estimate.
Suggested source: PVWatts study or solar proposal

Value / observation: _____

Source document, page, or person: _____

Electricity value [USD per kWh]

REQUIRED FOR MODEL

What to collect: The assumed value of each solar kWh, based on the applicable rate or credit treatment.
Why it matters: Converts production into dollar value. Utility tariff, meter arrangement, and credit rules can materially change it.
Suggested source: Utility bills, tariff, or documented modeling assumption

Value / observation: _____

Source document, page, or person: _____

Annual panel degradation [percent per year]

REQUIRED FOR MODEL

What to collect: Expected yearly decline in panel output.
Why it matters: Reduces generation in later years so lifetime value is not overstated.
Suggested source: Module warranty or production study

Value / observation: _____

Source document, page, or person: _____

Annual electricity-value escalation [percent per year]

REQUIRED FOR MODEL

What to collect: Expected annual change in the real value of electricity.
Why it matters: Changes the dollar value of future generation. Use zero when modeling in constant dollars without real escalation.
Suggested source: Documented analysis assumption

Value / observation: _____

Source document, page, or person: _____

Utility provider and rate class

PROJECT CONTEXT

What to collect: Name the utility, tariff, and account class if known.
Why it matters: Needed to verify the value assigned to solar electricity and whether credits can be allocated as assumed.
Suggested source: Recent utility bill

Value / observation: _____

Source document, page, or person: _____

Field data worksheet - continued

Meter configuration

PROJECT CONTEXT

What to collect: Describe master meters, separate tenant meters, common-area meters, and account holders.
Why it matters: Determines where solar production can offset usage and who can actually receive bill credits.
Suggested source: Single-line diagram, bills, or facilities team

Value / observation: _____

Source document, page, or person: _____

Who pays each electric bill?

PROJECT CONTEXT

What to collect: List the party responsible for each relevant meter or account.
Why it matters: Makes the value-allocation issue visible and helps the team design a structure that supports adoption instead of leaving savings stranded.
Suggested source: Utility accounts and lease documents

Value / observation: _____

Source document, page, or person: _____

ANALYSIS ASSUMPTIONS AND VALUE ALIGNMENT

These settings control the comparison period and show how usable solar value could reach project sponsors and building users.

Share of allocatable value reaching occupants / electricity users [percent]

REQUIRED FOR MODEL

What to collect: Starting scenario for the share of usable electricity value assigned to building users rather than the project/capital side.
Why it matters: Does not change generation; it changes the modeled distribution of value and helps identify shares where both modeled outcomes are positive.
Suggested source: Scenario assumption; not an agreement

Value / observation: _____

Source document, page, or person: _____

Value not currently assignable [percent]

REQUIRED FOR MODEL

What to collect: Share of electricity value that cannot yet be assigned to either modeled side.
Why it matters: Captures uncertainty, unusable credits, or unmatched load before the remaining value is divided.
Suggested source: Meter/utility review or conservative assumption

Value / observation: _____

Source document, page, or person: _____

Analysis horizon [years]

REQUIRED FOR MODEL

What to collect: Number of future years included in the comparison.
Why it matters: Controls how many years of production, maintenance, and planned roof activity are counted.
Suggested source: Project analysis convention

Value / observation: _____

Source document, page, or person: _____

Field data worksheet - continued

Discount rate [percent per year]

REQUIRED FOR MODEL

What to collect: Rate used to translate future project-side cash flows into today's dollars.
Why it matters: A higher rate gives less weight to distant savings and can change the preferred timing plan and NPV.
Suggested source: Organization's approved financial assumption

Value / observation: _____

Source document, page, or person: _____